

Traditional programming

↳ Input + Rules → Output

Machine Learning:-

Input + Historical data → Machine learns rule
↓
Predictions.

Arthur Samuel (1959)

the field of study that gives computers the ability to learn without being explicitly programmed.

Types
→ Supervised ML :- labeled data (T is present)
→ Unsupervised ML
→ Reinforcement ML.

Supervised ML

→ Collect labelled data
→ Train the ML model using this data
→ Test the model with new data
→ Use the trained model for prediction.
→ Classification:- Predicts categories or classes.
Ex:- disease diagnosis, spam det., image recognition.
→ Regression:- Predict continuous numerical values.
House price prediction, temperature forecasting

area	price
1200	45 lakh
1500	65 lakh
1800	85 lakh

Unsupervised Machine Learning no labeled data.

f1
f2
f3
f4
f5
f6
f7
f8
f9
f10
f11
f12
Similar → Group A → Sechana
Similar → Group B → Viragpur
Similar → Group C → Versicala

Types:-

→ Clustering → Grouping (k-means, Hierarchical clustering, DBSCAN)
→ Association rules → If X happens, Y is also likely to happen.
(EC2AT, FP-Growth)
→ Dimensionality reduction → PCA, t-SNE, UMAP, SVD

Reinforcement Learning

Agent → Action → Environment → Reward → Learn

ML Pipeline

→ flowchart

Problem → Collect data → Cleaning → EDA → Feature engineering
↓
Evaluation ← Training ← Model selection ← Train test split
↓
Deployment → Monitoring

100 dataset
↓
Train 80 Test 20
10 questions
9.8 ans
2 test